How do LLMs process prompts

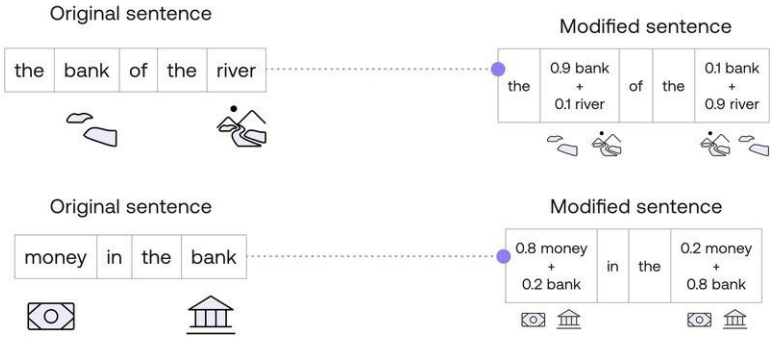
How to use Generative AI to create content

WARNING: This is a deeper dive

- If you aren't interested in getting a deeper understanding of LLMs, skip this lecture
- This goes deeper into the mechanisms of how LLMs understand human text and process it
- You don't need to understand this to use Generative AI effectively
- This is for people that want to learn about the encoder model and prompt processing

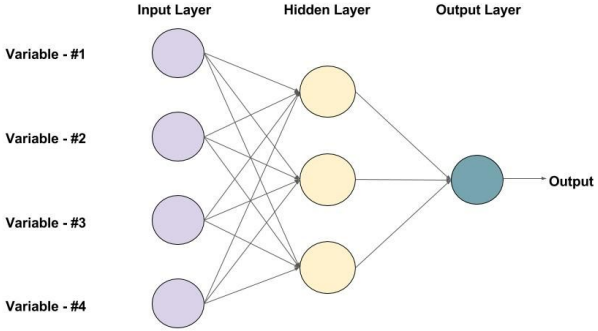
Attention is all you need

- Attention - A mathematical representation that allows LLMs to understand words used in different contexts.
    - Example: That t shirt is very cool vs It's cool outside. Cool is used in both sentences but have different meanings. Cool as in enthusiasm, and cool as in outside
- Google published a paper in 2017 called "Attention is all you need". Publicizing this a way of detecting words in certain contexts, consequently they created the transformer model
- Fun fact: this was originally meant for Google translate



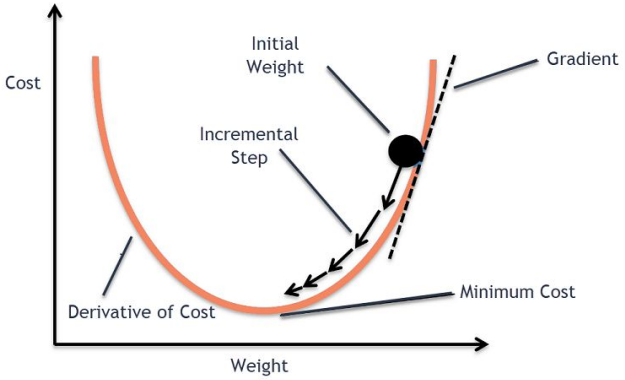
How does it work?

- Neural networks are a mathematical representation of how a human brain is structured with neurons being independent nodes connected together by edges with weights
- During training, the neural network is trained to reach a certain goal. The weights are randomly selected in the beginning and then in each iteration of training the weights are adjusted to try to get closer to the goal. This is called an epoch.



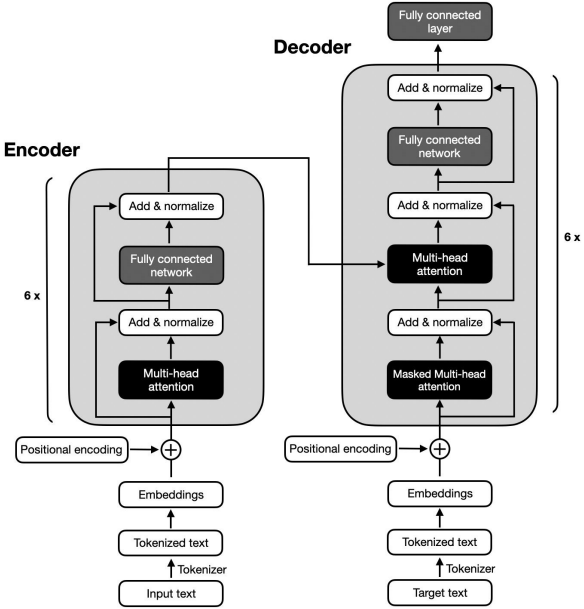
An example of a Feed-forward Neural Network with one hidden layer (with 3 neurons)

- A neural network might go through 1000 epochs, or 10000 epochs, or even 100000 epochs to reach a goal. Each epoch the weights are readjusted and the neural network attempts to get closer to the goal of identifying an object. This process is called training.
- When we get to the point where the neural network can't get more optimized then we reached a stopping point. The process of getting here is called gradient descent



Encoder and Decoders

- Encoders are neural networks for transforming text into a format that can be understood by the computer. This is called embedding and is fed into the decoder model



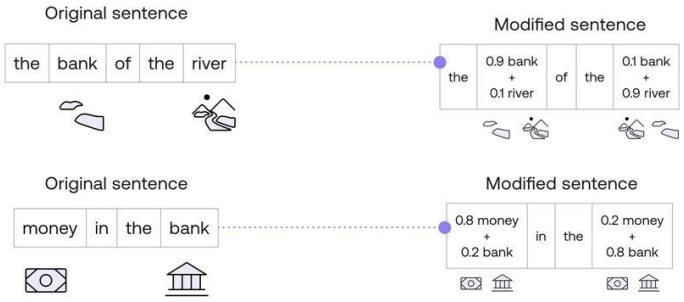
- Decoders are neural networks good at generating text and uses the input of encoding as attention to know where to begin outputting text
- Decoders are autoregressive, meaning that the output is fed back into the decoder until a special stop character is reached (remember transformers lecture)
- GPT is a decoder

How is this relevant to prompting?

- Computers don’t automatically understand language, or even letters.
- Computers understand numbers, and to get a computer to understand words you need to convert them into numbers.
- Embedding performs the conversion between text to a sparse vector that it can use to understand what you’re asking about

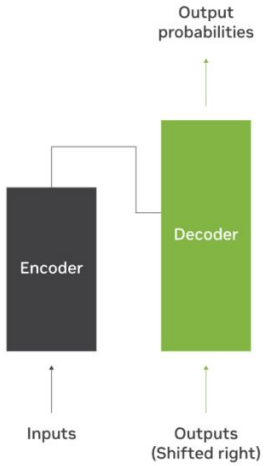
Back to attention

- During the embedding process we can adjust the weights based on the location and the words around it.
- During decoding, or generation, the computer works procedurally based on the query presented by the encoder
- Since we place different weights based on the sentences, the context can completely change the embedding vector making them unique

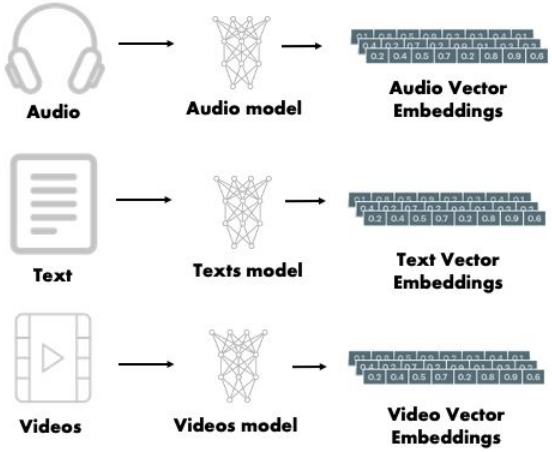


Putting it all together

- Now with a fully trained decoder model, and unique embeddings per prompt. The chatbot will be able to create unique responses for each input.
- This shows that prompting is important since placing more context allows you get a more specific tailored response.
- Adding more words entirely changes the embeddings into the LLMs that allows it access a different part of the knowledge base



Encoder-Decoders

Lets go multimodal

- The cool thing about embeddings is that they don't just work for text.
- You can convert audio, images, and more into embeddings that flow into a decoder model
- For those types of media you need to identify the important parts of the media and create weights for it.